Introduction to Computer Science: Homework 1

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Problem 1 — Is This Shuffle Random? (3 points)

Random shuffling appears in many places in computer science and everyday applications. For example, we may want to shuffle a deck of cards in a game, randomize the order of students in an experiment, or randomly permute data before training a machine learning model. In all these cases, we would usually like every possible ordering to be equally likely.

Consider the following algorithm for shuffling n elements, where $n \geq 3$.

Algorithm 1 Shuffling n elements

1: initially $\{a_k\} = \{k\}$, where $k \in \{1, 2, \ldots, n\}$
2: **for** $i = 1$ to n **do**
3: choose j uniformly from $\{1, 2, \ldots, n\}$
4: swap a_i and a_j
5: **end for**

Does this algorithm generate all $n!$ permutations with equal probability? Justify your answer.

Answer:

This algorithm does not generate all $n!$ permutations with equal probability.

There are n^n possible choice sequences in total, so the probability of each permutation is some integer times $1/n^n$. However, $n^n/n!$ is not an integer for $n \geq 3$: by Bertrand's postulate applied to $\lfloor n/2 \rfloor$, there always exists a prime p with $n/2 < p < n$ (for $n = 3$, take $p = 2$), which divides $n!$ but not n^n, since $p < n < 2p$. Hence $n! \nmid n^n$, and the distribution cannot be uniform.

Problem 2 — Transformer (3 points)

In this week's lecture, we discussed several representative achievements made possible by large language models (LLMs). Behind many of these systems is a fundamental architecture called the Transformer.

Transformer is such an important natural language processing model that even today's large models are still based on the Transformer architecture. The Transformer does not process raw text; it processes a sequence of tokens, where a token is just a small piece of text, such as a word or a punctuation mark.

(a) A core idea in Transformers is embedding: mapping discrete tokens to continuous vectors. Let $V = \{v_1, \ldots, v_n\}$ be a vocabulary of $n = 50000$ distinct tokens, and let each token v_i be a d-dimensional

vector. With $d = 768$, assume each token embedding is drawn uniformly from $[-1, 1]^d$ for simplicity. For any two distinct token embeddings v_i and v_j, what are the expectation and variance of

$$\cos(v_i, v_j) = \frac{v_i^\top v_j}{\|v_i\| \, \|v_j\|} \ ?$$

(b) Self-attention is a core component of the Transformer. For simplicity, we use a simplified version of it here.

When a passage of n tokens is fed into a Transformer, let its i-th token be represented by an embedding $x_i \in \mathbb{R}^d$. The layer has three learned matrices $Q, K, V \in \mathbb{R}^{d \times d}$, and

$$q_i = Qx_i, \qquad k_i = Kx_i, \qquad v_i = Vx_i.$$

For every pair i, j, define

$$c_{ij} = q_i \cdot k_j = \sum_{r=1}^{d} q_{i,r} k_{j,r}.$$

Then define

$$y_i = \sum_{j=1}^{n} c_{ij} v_j.$$

Define one operation as one multiplication of two real numbers or one addition of two real numbers. Compute how many operations are needed in total to obtain $y_1, \ldots, y_n$ from $x_1, \ldots, x_n$, and express the answer as a function of n and d.

(c) Suppose a Transformer has $L = 100$ layers, and each layer performs the computation described in part (b). On average, each query contains $n = 10^5$ tokens, with embedding dimension $d = 10^4$. Assume that each addition or multiplication consumes 2×10^{-10} J of energy. If 7×10^9 people each submit one such query per day, estimate the total energy consumed by the Transformer per day.

Answer:

For part (a), we define $u_i = v_i / \|v_i\|$ and $u_j = v_j / \|v_j\|$. Since normalizing a vector does not change the angle between two vectors, we have $\cos(v_i, v_j) = \cos(u_i, u_j)$. We define $u_{i,k}$ as the value of the k-th coordinate of u_i. Since v_i is drawn uniformly from the cube $[-1, 1]^d$, the distribution of u_i is invariant under flipping the sign of any single coordinate, and its coordinates are exchangeable. In particular, u_i and $-u_i$ have the same distribution.

Combining this with the fact that $\cos(-u_i, u_j) = -\cos(u_i, u_j)$, we obtain $\boxed{\mathbb{E}[\cos(v_i, v_j)] = \mathbb{E}[\cos(u_i, u_j)] = 0}$.

For the variance, since the expectation is 0, we have

$$\begin{aligned} \mathrm{Var}(\cos(u_i, u_j)) = \mathbb{E}[\cos^2(u_i, u_j)] &= \mathbb{E}\left[\sum_{p=1}^{d}\sum_{q=1}^{d} u_{i,p} u_{i,q} u_{j,p} u_{j,q}\right] \\ &= \sum_{p=1}^{d}\sum_{q=1}^{d} \mathbb{E}[u_{i,p} u_{i,q} u_{j,p} u_{j,q}], \end{aligned}$$

where the last equality follows from the linearity of expectation. For the cross terms $p \neq q$, since $u_{i,p}u_{i,q}$ and $u_{j,p}u_{j,q}$ are independent and since flipping the sign of the p-th coordinate gives $\mathbb{E}[u_{i,p}u_{i,q}] = -\mathbb{E}[u_{i,p}u_{i,q}] = 0$, we obtain $\mathbb{E}[u_{i,p}u_{i,q}u_{j,p}u_{j,q}] = \mathbb{E}[u_{i,p}u_{i,q}] \cdot \mathbb{E}[u_{j,p}u_{j,q}] = 0$. For the diagonal terms $p = q$, we have $\mathbb{E}[u_{i,p}^2 u_{j,p}^2] =$

$\mathbb{E}[u_{i,p}^2] \cdot \mathbb{E}[u_{j,p}^2]$. Since $\mathbb{E}[\|u_i\|^2] = \sum_{k=1}^{d} \mathbb{E}[u_{i,k}^2] = 1$ and the symmetry of u_i makes all the summands equal, we have $\mathbb{E}[u_{i,p}^2] = \mathbb{E}[u_{j,p}^2] = 1/d$.

Therefore, $\boxed{\mathrm{Var}(\cos(u_i, u_j)) = d \cdot \frac{1}{d^2} = \frac{1}{d} = \frac{1}{768}}$.

For part (b), computing all the q_i, k_i, v_i needs $3n(2d^2 - d)$ operations, computing all the c_{ij} needs $n^2(2d-1)$ operations and computing all the y_i needs $n(2n-1)d$ operations. In conclusion, obtaining $y_1, \ldots, y_n$ from $x_1, \ldots, x_n$ needs

$$3n(2d^2 - d) + n^2(2d-1) + n(2n-1)d = \boxed{6nd^2 + 4n^2d - n^2 - 4nd}$$

operations.

For part (c), one layer approximately needs $6 \times 10^5 \times 10^8 + 4 \times 10^{10} \times 10^4 \approx 4.6 \times 10^{14}$ operations. The answer can be estimated by

$$7 \times 10^9 \times 100 \times 4.6 \times 10^{14} \times 2 \times 10^{-10}\ \mathrm{J} \approx \boxed{6.4 \times 10^{16}\ \mathrm{J.}}$$

Problem 3 — Quantum Key Distribution (4 points)

In this week's lecture, we discussed quantum key distribution as a form of provable security based on the laws of quantum mechanics. In this problem, we will explore why eavesdropping can be detected.

Alice and Bob would like to establish a shared secret key using the BB84 quantum key distribution protocol. Unlike a classical bit, measuring a qubit may change its state.

A qubit can be represented by a normalized vector

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1,$$

where

$$|0\rangle = \begin{bmatrix}1\\0\end{bmatrix}, \qquad |1\rangle = \begin{bmatrix}0\\1\end{bmatrix}.$$

In this problem we use four states and two measurement bases:

$$|0\rangle\,, \quad |1\rangle\,, \quad |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}},$$

$$Z = \{|0\rangle\,, |1\rangle\}, \qquad X = \{|+\rangle\,, |-\rangle\}.$$

To measure a qubit in a basis, first express the state as a linear combination of the two basis states. If

$$|\psi\rangle = \alpha\,|b_0\rangle + \beta\,|b_1\rangle\,,$$

then measuring in the basis $\{|b_0\rangle\,, |b_1\rangle\}$ gives outcome $|b_0\rangle$ with probability $|\alpha|^2$ and $|b_1\rangle$ with probability $|\beta|^2$. After the measurement, the qubit is left in the state corresponding to the observed outcome.

3.1 Establishing a shared key

Alice chooses a random bit and independently chooses either the Z or X basis. She encodes the bit as

Classical bit	Z basis	X basis
0	$\|0\rangle$	$\|+\rangle$
1	$\|1\rangle$	$\|-\rangle$

Bob independently chooses either Z or X and measures the received qubit. Alice and Bob then publicly announce their bases and keep only the positions where their bases agree. This is called basis sifting.

(a) Suppose Alice and Bob choose between the X and Z bases independently and uniformly at random.

1) Show that, in the absence of noise and eavesdropping, Bob's classical bit is identical to Alice's bit at every retained position.
2) If Alice initially transmits N qubits, approximately how many shared random bits do Alice and Bob obtain?

Answer:

For part 1), on the retained positions the bases agree, so the state sent by Alice is one of the two orthogonal states of Bob's measurement basis, and expressing it in that basis gives coefficients 1 and 0. Hence Bob obtains the correct classical bit with probability 1.

For part 2), the bases agree with probability $1/2$ at each position, so approximately $N/2$ shared random bits are obtained.

The remaining shared bit string is called the sifted key.

3.2 What happens when Eve listens?

Suppose Eve intercepts every qubit. For each qubit, she independently chooses the X or Z basis uniformly at random, measures it, and sends the state she observed to Bob. This is called an intercept-resend attack.

For the following questions, consider only positions that Alice and Bob keep after basis sifting.

(b) Compute the expected error rate among the bits in Alice and Bob's sifted key.

(c) To detect eavesdropping, Alice and Bob randomly select m bits from their sifted key and publicly compare them. Assume that the communication channel is otherwise noiseless and that errors at different positions are independent. Using your result from part (b), derive the probability that Eve is detected. What happens to this probability as m increases?

Answer:

For part (b), an error can be caused only by the situation that Eve chooses the wrong basis, in which case Bob's bit is wrong with probability 1/2. The probability of this situation is $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, so the error rate is $\frac{1}{4}$.

For part (c), the probability of detecting Eve is $1-(1-1/4)^m = 1-(3/4)^m$; as m increases, this probability approaches 1.

Declaration of AI Use

I used an AI assistant on all problems in this homework, for grammar polishing, LaTeX typesetting, and checking the correctness of my arguments. All mathematical ideas and proofs are my own.